

Section 4.5 — Business Transformation Documentation

1. Purpose and Overview

The Business Transformation layer is the fourth stage in the ETL pipeline (Source → Staging → Validation → Business Transformation → Data Warehouse). It is implemented as a dedicated SSIS package (Transformation.dtsx) that reads exclusively from the _Valid output of the Validation layer — no _Invalid records are ever processed here, per Requirement 5 (Valid Records Processing) established in the Validation layer.

The layer processes all 14 tables that were validated in the previous stage — 4 Fact tables (FactSales, FactOnlineSales, FactInventory, FactExchangeRate) and 10 Dimension tables — and applies three categories of business logic to each, as required by the project brief (Section 3.4):

1. **Financial Calculations** — applied to the 4 Fact tables, which contain the transactional/measurable data.
2. **Business Classifications** — applied wherever a table contains an attribute that can be meaningfully grouped or categorized (both Facts and Dimensions).
3. **Data Standardization** — applied to text fields across Dimension tables (Proper Case, trimming, phone number formatting).

A fourth component, **Audit Information**, is applied uniformly to all 14 tables regardless of category.

2. Architecture

Source: validation.<Table>_Valid (14 tables)



Derived Column: Financial Calculations (Facts only)



Derived Column: Business Classification (where applicable)



Derived Column: Data Standardization (Dimensions, text fields)



Derived Column: Audit Information (all 14 tables)



Destination: transformation.<Table>_Transformed (14 tables)

The Control Flow separates Fact and Dimension processing into two Sequence Containers (Facts_Transformation and Dims_Transformation), preceded by a single Execute SQL Task that truncates all 14 transformation schema tables to ensure idempotent re-execution on each run.

3. Financial Calculations (Fact Tables)

3.1 FactSales

Column	Formula	Business Purpose
GrossProfit	SalesAmount - TotalCost	Gross margin before discounts/returns
NetSalesAmount	SalesAmount - DiscountAmount - ReturnAmount	Actual net revenue after adjustments
ProfitMarginPercent	$(\text{SalesAmount} \neq 0) ? ((\text{SalesAmount} - \text{TotalCost}) / \text{SalesAmount}) * 100 : 0$	Profitability ratio, protected against division by zero
AverageUnitProfit	$(\text{SalesQuantity} \neq 0) ? (\text{SalesAmount} - \text{TotalCost}) / \text{SalesQuantity} : 0$	Per-unit profitability, protected against division by zero

3.2 FactOnlineSales

Same four calculations as FactSales (GrossProfit, NetSalesAmount, ProfitMarginPercent, AverageUnitProfit), applied identically for consistency across both sales channels, using the equivalent columns available in the FactOnlineSales schema.

3.3 FactInventory

FactInventory does not contain a SalesAmount column, so calculations are based on stock valuation instead:

Column	Formula	Business Purpose
InventoryValue	OnHandQuantity * UnitCost	Monetary value of stock currently on hand
OnOrderValue	OnOrderQuantity * UnitCost	Monetary value of stock on order
StockCoverageDays	$(\text{!ISNULL}(\text{SafetyStockQuantity}) \ \&\& \ \text{SafetyStockQuantity} \neq 0) ? \text{OnHandQuantity} / \text{SafetyStockQuantity} : 0$	Ratio of current stock to safety stock threshold

3.4 FactExchangeRate

FactExchangeRate contains only currency rate data (no sales/cost figures), so calculations focus on rate volatility:

Column	Formula	Business Purpose
RateVariance	EndOfDayRate - AverageRate	Absolute intraday rate movement

Column	Formula	Business Purpose
RateVariancePercent	$(\text{AverageRate} \neq 0) ? ((\text{EndOfDayRate} - \text{AverageRate}) / \text{AverageRate}) * 100 : 0$	Percentage intraday rate movement

4. Business Classifications

Classifications are distributed across whichever table holds the relevant attribute, not concentrated only in the Fact tables:

Table	Column	Logic	Purpose
FactInventory	StockStatus	Out of Stock / Below Safety Stock / In Stock (based on OnHandQuantity vs SafetyStockQuantity)	Operational stock alerting
FactExchangeRate	RateTrend	Appreciating / Depreciating / Stable (based on EndOfDayRate vs AverageRate)	Currency movement summary
DimPromotion	PromotionStatus	Expired / Upcoming / Active (based on StartDate/EndDate vs current date)	Promotion lifecycle tracking
DimPromotion	DiscountTier	High / Medium / Low / No Discount (based on DiscountPercent)	Promotion strength segmentation
DimStore	StoreOperationalStatus	Operating / Closed (based on CloseDate)	Store lifecycle tracking
DimStore	StoreSizeCategory	Large / Medium / Small / Unknown (based on SellingAreaSize)	Store footprint segmentation
DimProduct	PriceTier	Premium / Mid-Range / Budget / Unknown (based on UnitPrice)	Product pricing segmentation
DimProduct	ProductAvailability	Available / Discontinued (based on StopSaleDate)	Product lifecycle tracking
DimCustomer	IncomeSegment	High / Medium / Low Income (based on YearlyIncome)	Customer income segmentation
DimCustomer	CustomerLifeStage	Family / Individual (based on TotalChildren)	Customer household segmentation

Three tables were intentionally **excluded** from classification, as no attribute in their schema supports a meaningful business grouping: DimCurrency, DimChannel, and DimDate (the latter already being a fully

pre-built calendar dimension). DimGeography, DimProductCategory, and DimProductSubcategory similarly received no additional classification, as their columns are themselves already categorical in nature.

5. Data Standardization

Applied to text-bearing Dimension columns, following the project brief's explicit requirement (Section 3.4): Proper Case for names, standardized phone numbers, trimmed spaces, NULL handling.

Table	Standardization Applied
DimCurrency	CurrencyName_Clean — uppercase, trimmed
DimChannel	ChannelLabel_Clean — uppercase, trimmed
DimGeography	CityName_Clean, RegionCountryName_Clean — uppercase, trimmed, NULL-safe
DimProductCategory	ProductCategoryName_Clean — uppercase, trimmed
DimProductSubcategory	ProductSubcategoryName_Clean — uppercase, trimmed
DimPromotion	PromotionName_Clean — uppercase, trimmed
DimStore	StoreName_Clean (uppercase, trimmed), StorePhone_Clean (dashes/parentheses removed)
DimProduct	ProductName_Clean, BrandName_Clean — trimmed, NULL-safe (kept in original case, as product names are typically displayed in their native casing rather than forced uppercase)
DimCustomer	FirstName_Clean, LastName_Clean — true Proper Case (first letter capitalized, remainder lowercase); Phone_Clean — dashes/parentheses removed

DimDate received no standardization, as it is a fully generated reference dimension with no free-text fields requiring cleanup.

6. Audit Information

Per Section 3.4 of the project brief, every one of the 14 transformed tables includes a uniform audit trail, appended as the final Derived Column step before loading:

Column	Source	Purpose
LoadDate	GETDATE()	Timestamp of when the row was processed
PackageName	@[System::PackageName]	SSIS system variable — records which package produced the row
ETLUser	@[System::UserName]	SSIS system variable — records which OS user executed the package
BatchID	@[User::BatchID]	Package-scoped variable, evaluated once per execution via the expression (DT_WSTR,50)@[System::ExecutionInstanceGUID], ensuring every row loaded in the same run shares one identifying value

The BatchID variable is deliberately evaluated **once per package execution** (not once per row) using System::ExecutionInstanceGUID, so that all rows loaded in a single run — whether via Full Load or Incremental Load — can be traced back to that specific execution as a group, supporting rollback and audit scenarios (e.g., DELETE FROM ... WHERE BatchID = '<value>' to undo a single load).

7. Destination Tables

All 14 transformed tables are written to a dedicated transformation schema, following the naming convention transformation.<TableName>_Transformed:

Facts: FactSales, FactOnlineSales, FactInventory, FactExchangeRate **Dimensions:** DimCurrency, DimChannel, DimGeography, DimProductCategory, DimProductSubcategory, DimPromotion, DimDate, DimStore, DimProduct, DimCustomer

8. Implementation Notes and Challenges

- All numeric calculations involving division (ProfitMarginPercent, AverageUnitProfit, RateVariancePercent, StockCoverageDays) include explicit divide-by-zero protection using the SSIS ternary conditional operator, since the source data occasionally contains zero-value denominators.
- DimDate was treated as a special case, receiving only the Audit Information step, since it is a pre-generated reference dimension with no free-text or classifiable business attributes.
- As with the Staging and Validation layers, each OLE DB Destination table had to be executed once individually (via "New..." table auto-generation) before it could be referenced in the shared Truncate Execute SQL Task — omitting this step for any single table produces a "Cannot find the object" error at Truncate time, as encountered during development for DimGeography_Transformed.